

VC_RANDOM

Overview

vc_random package contains the set of useful randomization functions.

Functions		
Name	Arguments	Description
rn	Min: integer Max: integer	Returns the random value between Min and Max (inclusive).
rl	Min: integer Max: integer	Returns the random value between Min and Max (inclusive), with the more probablitiy given for the small numbers
rh	Min: integer Max: integer	Returns the random value between Min and Max (inclusive), with the more probablitiy given for the large numbers
mnc2 to mnc5	Value1, weight1; Value2, weight2	Returns one of the specified input values. Uses Input values random weights to perform constraint-random choice.

Example1

```
module test ();  
  
reg [0:31] seed;  
integer i;  
  
`include "vc_random.v"  
  
initial begin  
    rand.seed = 10;  
    for (i=0; i<40; i=i+1)  
        $display("rl = %0d    rn = %0d    rh = %0d",  
            rl(0,20), rn(0,20), rh(0,20));  
    $finish;  
end  
  
endmodule
```

Example2: Constraint-random FSM

```
module sender (clk, rst, b2s_ack, s2b_req);
input clk, rst;
input b2s_ack;
output s2b_req;

parameter IDLE = 0;
parameter BUSY = 1;
reg state;

`include "vc_random.v"

always @(posedge clk) begin
    if (rst) state <= IDLE;
    else begin
        if (state == IDLE && !b2s_ack) state <= rnc2(IDLE, 10-`SND_SPEED, BUSY,
`SND_SPEED);
        if (state == BUSY && b2s_ack) state <= rnc2(BUSY, 10-`SND_SPEED, IDLE,
`SND_SPEED);
    end
end

assign s2b_req = (state == BUSY);

endmodule
```